

YOUR PERSONALIZED REPORT

PATHOGENS

The testing includes pathogens (bacterial, parasitic and viral) commonly known to cause gastroenteritis. Note that not all individuals with positive findings will present with symptoms. Many factors, including the health of the individual (such as immune health, digestive function, and microbiome balance), the transient nature of most pathogens, and the presence and expression of virulence factors, all contribute to pathogen virulence and individual symptoms.

BACTERIAL PATHOGENS

	Result	Reference
<i>Campylobacter</i>	<dl	< 1.00e3
<i>C. difficile</i> Toxin A	<dl	< 1.00e3
<i>C. difficile</i> Toxin B	<dl	< 1.00e3
<i>Enterohemorrhagic E. coli</i>	<dl	< 1.00e3
<i>E. coli</i> O157	<dl	< 1.00e3
Enteroinvasive <i>E. coli</i> / <i>Shigella</i>	<dl	< 1.00e3
Enterotoxigenic <i>E. coli</i> LT/ST	<dl	< 1.00e3
Shiga-like Toxin <i>E. coli</i> stx1	<dl	< 1.00e3
Shiga-like Toxin <i>E. coli</i> stx2	<dl	< 1.00e3
<i>Salmonella</i>	<dl	< 1.00e4
<i>Vibrio cholerae</i>	<dl	< 1.00e5
<i>Yersinia enterocolitica</i>	<dl	< 1.00e5

PARASITIC PATHOGENS

<i>Cryptosporidium</i>	<dl	< 1.00e6
<i>Entamoeba histolytica</i>	<dl	< 1.00e4
<i>Giardia</i>	<dl	< 5.00e3

VIRAL PATHOGENS

Adenovirus 40/41	<dl	< 1.00e10
Norovirus GI/II	<dl	< 1.00e7

KEY: Results are reported as genome equivalents per gram of stool, which is a standard method for estimating the number of microbes measured per gram of stool, based on qPCR analysis of DNA samples.

Results are expressed in standard scientific notation. For example, a reported result of 3.5e7 is equivalent to 3.5×10^7 microbes per gram, which equals 35,000,000 (35 million) microbes per gram of stool.

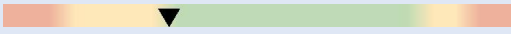
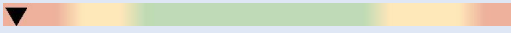
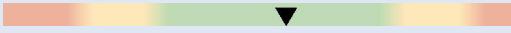
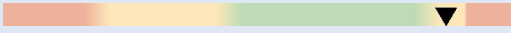
< dl represents results below detectable limit.

HELICOBACTER PYLORI

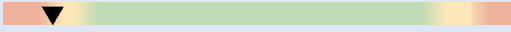
H. PYLORI & VIRULENCE FACTORS

	Result	Reference
<i>Helicobacter pylori</i>	1.42e3 High ↑	< 1.00e3
Virulence Factor, babA	Negative	Negative
Virulence Factor, cagA	Negative	Negative
Virulence Factor, dupA	Negative	Negative
Virulence Factor, iceA	Negative	Negative
Virulence Factor, oipA	Negative	Negative
Virulence Factor, vacA	Negative	Negative
Virulence Factor, virB	Negative	Negative
Virulence Factor, virD	Negative	Negative

COMMENSAL/KEYSTONE BACTERIA

COMMENSAL BACTERIA	Result	Reference
<i>Bacteroides fragilis</i>	6.03e9 	1.6e9 - 2.5e11
<i>Bifidobacterium</i> spp.	1.14e10 	> 6.7e7
<i>Enterococcus</i> spp.	7.29e4 L 	1.9e5 - 2.0e8
<i>Escherichia</i> spp.	1.05e6 L 	3.7e6 - 3.8e9
<i>Lactobacillus</i> spp.	2.73e7 	8.6e5 - 6.2e8
<i>Enterobacter</i> spp.	5.47e6 	1.0e6 - 5.0e7
<i>Akkermansia muciniphila</i>	3.73e6 	1.0e1 - 8.2e6
<i>Faecalibacterium prausnitzii</i>	8.88e2 L 	1.0e3 - 5.0e8
<i>Roseburia</i> spp.	6.35e8 	5.0e7 - 2.0e10

BACTERIAL PHYLA

<i>Bacteroidetes</i>	1.25e12 	8.6e11 - 3.3e12
<i>Firmicutes</i>	5.71e10 	5.7e10 - 3.0e11
<i>Firmicutes:Bacteroidetes</i> Ratio	0.05 	< 1.0

OPPORTUNISTIC/OVERGROWTH MICROBES

DYSBIOTIC & OVERGROWTH BACTERIA	Result	Reference
<i>Bacillus</i> spp.	8.72e5	< 1.76e6
<i>Enterococcus faecalis</i>	<dl	< 1.00e4
<i>Enterococcus faecium</i>	2.43e3	< 1.00e4
<i>Morganella</i> spp.	<dl	< 1.00e3
<i>Pseudomonas</i> spp.	<dl	< 1.00e4
<i>Pseudomonas aeruginosa</i>	8.31e2 High ↑	< 5.00e2
<i>Staphylococcus</i> spp.	<dl	< 1.00e4
<i>Staphylococcus aureus</i>	1.20e3 High ↑	< 5.00e2
<i>Streptococcus</i> spp.	1.24e4 High ↑	< 1.00e3
COMMENSAL OVERGROWTH MICROBES		
<i>Desulfovibrio</i> spp.	5.55e7	< 7.98e8
<i>Methanobacteriaceae</i> (family)	1.08e8	< 3.38e8
INFLAMMATORY & AUTOIMMUNE-RELATED BACTERIA		
<i>Citrobacter</i> spp.	<dl	< 5.00e6
<i>Citrobacter freundii</i>	<dl	< 5.00e5
<i>Klebsiella</i> spp.	<dl	< 5.00e3
<i>Klebsiella pneumoniae</i>	<dl	< 5.00e4
<i>M. avium</i> subsp. <i>paratuberculosis</i>	<dl	< 5.00e3
<i>Proteus</i> spp.	<dl	< 5.00e4
<i>Proteus mirabilis</i>	<dl	< 1.00e3
COMMENSAL INFLAMMATORY & AUTOIMMUNE-RELATED BACTERIA		
<i>Enterobacter</i> spp.	5.47e6	< 5.00e7
<i>Escherichia</i> spp.	1.05e6	< 3.80e9
<i>Fusobacterium</i> spp.	9.48e5	< 1.00e8
<i>Prevotella</i> spp.	1.84e7	< 1.00e8

FUNGI/YEAST

FUNGI/YEAST	Result	Reference
<i>Candida</i> spp.	<dl	< 5.00e3
<i>Candida albicans</i>	<dl	< 5.00e2
<i>Geotrichum</i> spp.	<dl	< 3.00e2
<i>Microsporidium</i> spp.	<dl	< 5.00e3
<i>Rhodotorula</i> spp.	<dl	< 1.00e3

VIRUSES

VIRUSES	Result	Reference
Cytomegalovirus	<dl	< 1.00e5
Epstein-Barr Virus	<dl	< 1.00e7

PARASITES

PROTOZOA	Result	Reference
<i>Blastocystis hominis</i>	<dl	< 2.00e3
<i>Chilomastix mesnili</i>	<dl	< 1.00e5
<i>Cyclospora</i> spp.	<dl	< 5.00e4
<i>Dientamoeba fragilis</i>	<dl	< 1.00e5
<i>Endolimax nana</i>	1.77e3	< 1.00e4
<i>Entamoeba coli</i>	<dl	< 5.00e6
<i>Pentatrichomonas hominis</i>	<dl	< 1.00e2
WORMS		
<i>Ancylostoma duodenale</i>	Not Detected	Not Detected
<i>Ascaris lumbricoides</i>	Not Detected	Not Detected
<i>Necator americanus</i>	Not Detected	Not Detected
<i>Trichuris trichiura</i>	Not Detected	Not Detected
<i>Taenia</i> spp.	Not Detected	Not Detected

INTESTINAL HEALTH MARKERS

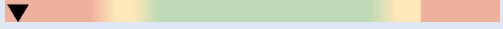
DIGESTION

	Result	Reference
Steatocrit	<dl 	< 15 %
Elastase-1	>750 	> 200 ug/g

GI MARKERS

β-Glucuronidase	1395 	< 2486 U/mL
Occult Blood - FIT	67 H 	< 10 ug/g

IMMUNE RESPONSE

Secretory IgA	239 L 	510 - 2010 ug/g
Anti-gliadin IgA	>500 H 	< 175 U/L
Eosinophil Activation Protein (EDN, EPX)	0.09 	< 2.34 ug/g

INFLAMMATION

Calprotectin	66 	< 173 ug/g
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H. PYLORI ANTIBIOTIC RESISTANCE GENES

	Result	Reference
Amoxicillin	Negative	Negative
<i>Genes associated with amoxicillin resistance</i>		
PBP1A S414R	Absent	
PBP1A T556S	Absent	
PBP1A N562Y	Absent	

	Result	Reference
Clarithromycin	Negative	Negative
<i>Genes associated with clarithromycin resistance</i>		
A2142C	Absent	
A2142G	Absent	
A2143G	Absent	

	Result	Reference
Fluoroquinolones	Negative	Negative
<i>Genes associated with fluoroquinolone resistance</i>		
gyrA N87K	Absent	
gyrA D91N	Absent	
gyrA D91G	Absent	
gyrB S479N	Absent	
gyrB R484K	Absent	

	Result	Reference
Tetracycline	Negative	Negative
<i>Genes associated with tetracycline resistance</i>		
A926G	Absent	
AGA926-928TTC	Absent	